# Promise Osaine Ekpo

Human Robot Collaboration and Grounded Robot Communication

**Ph.D. graduation December 2027.** Available for a 12 week internship, Summer 2027.
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## Research Focus

I build datasets, benchmarks, and learning systems for robots that collaborate with people. My research models human knowledge and responses to robot help requests, predicts interaction success, and studies multimodal failure recovery. I combine controlled human studies, physical robot deployments, MuJoCo simulation, and reinforcement learning to investigate human-aware robot behavior and transfer across robot embodiments.

## Education

**Cornell University, Ph.D. Computer Science.** Advisor Prof. Angelique Taylor. GPA 3.94/4.00. *Aug 2023 to Dec 2027 expected*
**Princeton University, M.S.E. Computer Science.** Advisor Prof. Jaime Fernández Fisac. GPA 3.67/4.00. *Aug 2021 to May 2023*
**University of Benin, B.Eng. Computer Engineering.** GPA 4.77/5.00. Best Graduating Student, Dean's Honor Award. *Sep 2015 to Mar 2020*

## Publications

Google Scholar

**Peer reviewed and accepted**

- Batra, Y., Pioldi, G., **Ekpo, P.**, et al. RFM HRI: A Multimodal Dataset of Medical Robot Failure, User Reaction, and Recovery Preferences. Accepted, *ACM Transactions on Human Robot Interaction*, 2026.
- Tanjim, T., **Ekpo, P.**, Cao, H., et al. Human Robot Teaming Field Deployments: A Comparison Between Verbal and Nonverbal Communication. *IEEE RO-MAN*, 2025, pp. 1699–1704.
- **Ekpo, P.**, Taylor, A., Molu, L. A Generalized Nash Equilibrium Seeking Scheme for Trauma Resuscitation. *IFAC World Congress*, 2026.
- Ogundare, O., Owadokun, T., Ogundare, T., **Ekpo, P.**, et al. Integrated Artificial Intelligence in Healthcare and the Patient's Experience of Care. *Scientific Reports*, 2025.
- Ogundare, O., **Ekpo, P.**, Wiggins, N. Ladder Logic Translation using Large Language Models in Industrial Automation. Accepted, *INFUS 2026* (Springer proceedings).
- **Ekpo, P.**, Agarwal, S., Grimm, F., Molu, L., Taylor, A. AdaFair MARL: Enforcing Adaptive Fairness Constraints in Multi Agent Reinforcement Learning. Poster accepted, *Women in Machine Learning Workshop at NeurIPS*, 2026.

**Manuscripts and preprints**

- **Ekpo, P.**, Vijay, T., Mandalik, D., Jain, T., Ibrayeva, A., Sil, S., Tellex, S., Taylor, A. Toward Human-in-the-Loop Robot Failure Recovery: Bridging Communication Gaps in Human-Robot Collaboration. Under review, *ACM CHI 2027*.
- **Ekpo, P.**, Jain, T., Sayatqyzy, A., Sil, S., Taylor, A. Toward Reducing Knowledge Induced Performance Gaps with Expertise Conditioned Inverse Semantics. Under review, *ACM/IEEE HRI 2027*.
- Pioldi, G., Batra, Y., Bai, Y., **Ekpo, P.**, Taylor, A. REPAIR Bench: A Benchmark for Robot Error Perception and Interaction Recovery. Preprint, 2026.
- **Ekpo, P.**, La, B., Wiener, T., et al. Skill Aligned Fairness in Multi Agent Learning for Collaboration in Healthcare. Preprint, 2025.

## Selected Research

**Listener Differences in Human Robot Interaction, a Game, Dataset, and Benchmark** 2025 to Present
*AIRLab, Cornell Tech*

- Built LD HRI, a framework, game, dataset, and benchmark for when the same robot help request succeeds for one person and fails for another, across four tasks varying control interface, robot part, and scene familiarity.
- Collected 240 completed study sessions and analyzed 255 clean matched Expert and Novice request pairs. Identical human authored requests yielded 80.0% versus 64.3% success, a 15.7 percentage point gap, exact McNemar $p < .001$.
- Built the web simulator, matched message delivery, eight trial configurations, PostgreSQL logging, and PDDL reconstructable event traces. Developed benchmarks for language model Speakers, Listeners, and request selection.

**Robot Failure Detection and Interaction Recovery** 2024 to Present
*AIRLab, Cornell Tech*

- Developed RFM HRI and contributed to REPAIR Bench using 214 physically grounded trials with 41 participants to model robot failures, user reactions, and preferred recovery strategies.
- Built multimodal failure detection and temporal localization models, achieving strict failure F1 of 0.80 versus 0.68 and median localization error of 2.97 seconds. Fine tuned Mistral 7B with QLoRA for recovery preference prediction, achieving Hit@5 of 0.76.

## Selected Research (continued)

**Expertise Conditioned Inverse Semantics for Robot Help Requests** 2026 to Present
*AIRLab, Cornell Tech*

- Extended inverse semantics with an explicit latent variable for task relevant listener knowledge, online belief updates from observed behavior, and a communication cost that supplies the mapping a Novice lacks while avoiding information an Expert already holds.
- Froze the model specification on 272 outcomes from 49 listeners, then evaluated it without refitting on 80 later collected trials from 10 entirely new listeners and a previously unseen maintenance task. Frozen context aware EIV reached 0.655 area under the receiver operating characteristic curve, 0.228 Brier score, 0.688 accuracy, and 0.576 macro F1, against 0.458 curve area for an utterance and task baseline. The paired listener bootstrap improvement was 0.192 with a 95 percent interval of 0.028 to 0.320.
- Validating the algorithm on real world deployment tasks with a Kinova arm and a TurtleBot 4, so that selected requests are evaluated in the real world with physical hardware.
- Reached 0.709 curve area on a developmental rerun over all 352 outcomes with a context plus task by utterance model, and reported the limits plainly. The results support held out prediction on later listeners but do not yet show that the policy closes the human gap.
- Introduced an evaluator that checks whether a task's features can express the distinctions a listener needs before any listener model is fitted, after finding that room names and turn counts cannot represent route layout or listener heading.

**Cross Embodiment Transfer Based on Human Understanding of Robot Capabilities** 2026 to Present
*AIRLab, Cornell Tech*

- Built a MuJoCo benchmark for robot failure recovery with simulated robot arms and programmed human responses. A software fault blocks motor commands until a helper resets the robot. Compared general help requests, specific reset instructions, and explanations followed by instructions.
- Developed response models combining task context, request wording, robot hardware features, and individual capability beliefs. Coupled predicted helping probabilities with simulated physical outcomes to select requests maximizing expected task completion while keeping the movement controller fixed.
- Evaluated four transfer protocols across 20 fixed seeds, covering disjoint sets of three versus six robot embodiments and three-joint versus six-joint arms in both directions. Separated robot embodiments, simulated people, and physical scene banks between training and testing, with balanced source-data budgets.
- On held-out six-joint arms after training on three-joint arms, adding capability beliefs reduced Brier prediction error from 23.15 to 19.02 on a 0–100 scale. The paired reduction was 4.14 points, with a 95% bootstrap interval of [3.94, 4.35]. Request selection increased expected task completion from 68.36% to 71.02% under the shared programmed response rule.
- Compared transfer without adaptation, limited target-data adaptation, target-native training, and ablations of hardware and belief inputs. Tested 14 fixed sensitivity settings and wrong-person belief profiles. Reversing the belief-response relationship on target robot embodiments harmed transfer, identifying a concrete limit of the model.
- Designing studies linking measured capability beliefs to observed helping actions across TurtleBot 4, Reachy Mini, a robotic crash cart, and TrashBot. Prepared synchronized interaction, action, language, and perception collection with robot hardware and Vicon tracking. Current transfer results concern human-response prediction in simulation, not learned motor-policy transfer or demonstrated gains with human participants.

**Human Aware Social Navigation and Robot Validation** 2026 to Present
*AIRLab, Cornell Tech*

- Built a human aware navigation stack combining pedestrian trajectory prediction, model predictive control, and control barrier function safety filters, and validated the resulting behavior on TurtleBot 4 hardware.
- Integrated the Create 3 base, RPLIDAR, OAK D Pro RGB D camera, inertial measurement unit, and wheel encoders through ROS 2 Humble on Linux, and achieved repeatable point to point navigation by generating SLAM maps, launching map based localization, setting poses in RViz, executing Nav2 goals, and debugging LiDAR and odometry failures.

**Vision Language Models (VLMs) as Inverse Optimization Designers** 2026 to Present
*AIRLab, Cornell Tech*

- Reduced navigation weight recovery error from 0.190 to 0.004 using 20 demonstrations, against 0.057 for maximum entropy inverse reinforcement learning, by implementing active set inverse optimization under a simplex constraint.
- Showed with two controlled tests that the VLM was copying its in context examples instead of reasoning about the scene. Opposing examples moved the recovered weights apart by 1.116, while changing the scene with examples held fixed moved them by zero.
- Traced the failure to the weight estimator using the NavWareSet dataset and evaluation pipeline over 12 scenes, six matched socially aware and socially unaware navigation pairs, 466 robot person encounters, and 33 checks.

**Robotic Crash Cart Communication in Clinical Simulations** 2024 to 2025
*AIRLab, Cornell Tech*

- Supported field deployment and evaluation with 115 healthcare workers at Weill Cornell Medicine, comparing verbal, visual, and traditional cart conditions using NASA Task Load Index and workplace attitude measures.
- Found lower operator effort for speech than visual cues and the traditional cart, with $p = .021$ and $p = .019$, respectively, informing communication design for healthcare teamwork.

**MARLHospital Simulator and Skill Aligned Fairness** 2023 to Present
*AIRLab, Cornell Tech*

- Built a simulator for heterogeneous team coordination, fatigue, and shared tasks. Implemented QMIX, MAPPO, VDN, IQL, and IPPO in PyTorch, scaling reproducible evaluation to more than 200 concurrent multi GPU Slurm jobs.
- Replaced hand tuned penalties with Lagrangian dual ascent, improving Jain's fairness index from 0.33 to 0.89 at 86 percent task success.

## Additional Research

**Safe and Personalized Autonomy Research** 2021 to 2023
*Safe Robotics lab, Princeton University*

- Conducted master's thesis research in Prof. Jaime Fernández Fisac's Safe Robotics Lab using a game theoretic framework to measure persuasiveness and safety risks in GPT 3 and GPT 4, measuring belief shifts of 12 to 50 percent toward false statements, and trained language models with proximal policy optimization over 5,000 iterations.
- In Prof. Barbara Engelhardt's Beehive Research Laboratory, used multi group Gaussian processes and fitted Q iteration to study personalized healthcare policies across patient populations.

## Industry Experience

**Multimodal Safety Monitor and Explainable AI, Project Titanium** Summer 2023
*Siemens, Robust Intelligence Team*

- Built a Python warehouse safety monitor that fused vision based detection with a real time location system so that each sensing stream could cross check the other around robots and people.
- Applied GNNExplainer to graph attention networks for interpretable medical diagnosis, and presented both systems at the final review.

**Data Engineer** 2020 to 2021
*Infinion Technologies, Microsoft Gold Partner*

- Contributed to the Union Bank data warehouse migration, designed data standards, and built production ETL processes with SQL Server Integration Services and SQL.

## Technical Skills

**Robotics and Autonomy.** ROS 2 Humble, TurtleBot 4, Create 3 base, Reachy Mini, Kinova arm, SLAM, mapping, localization, Nav2, RViz, teleoperation, RPLIDAR, OAK D Pro RGB D camera, inertial measurement unit, wheel encoders, Vicon motion capture
**Simulation and Validation.** MuJoCo setup and debugging, custom multi agent simulators, physics simulation, automated validation frameworks, dataset and experiment pipelines
**Learning and Perception.** PyTorch, TensorFlow, JAX, Keras, reinforcement learning, multi agent reinforcement learning, inverse optimization, maximum entropy inverse reinforcement learning, inverse semantics, PDDL, vision language model evaluation, QLoRA and parameter efficient fine tuning, image processing, video segmentation
**Programming and Systems.** Python, Linux, Git, SQL, PostgreSQL, Slurm, multi GPU training and profiling, cloud deployment

## Selected Talks

**Human Robot Teaming Field Deployments: A Comparison Between Verbal and Nonverbal Communication.** 2025
Princeton Robotics Colloquium
**MARLHospital, a multi agent reinforcement learning environment for medical decision making.** Tapia Conference 2025
**A Generalized Nash Equilibrium Seeking Scheme for Trauma Resuscitation.** IFAC World Congress poster 2026
**MARLHospital and Skill Aligned Fairness.** NCWIT Summit, Bank of America luncheon 2026

## Honors and Awards

IEEE Robotics and Automation Society SPRISE Competition, Third Place, IROS 2026
Grace Hopper Celebration, Emerging Technologist Award, Winner 2026
NCWIT Aspirations in Computing Collegiate Award, Grand Prize ($8,500) 2026
CMD IT and ACM Richard Tapia Conference, First Place Graduate Research Poster ($1,500) 2025
Bowers CIS Inspiring Futures Award for mentorship 2025
Cornell Graduate School Exemplary Leadership Award for Early Career Graduate Students 2024
Cadence Technology Award, Women in Technology ($5,000) 2023

## Service and Teaching

- **Mentoring.** 18 undergraduate, master's, and Ph.D. researchers across HRI, robot learning, language models, multi agent learning, and hardware study operations.
- **Workshop preparation and outreach.** Contributed to an HRI 2027 workshop proposal on long term real world robot deployments, including external relations and panelist outreach; proposal not submitted. Co-organized the Black in Robotics session at IROS 2026 Community Building Day.
- **Outreach.** Black in Robotics workshops, 2023 and 2025 to present. Research speaker and robot demonstrator, Women in Robotics event at NYC FIRST Robotics, 2026.
- **Reviewer.** ACM/IEEE HRI, ACM Transactions on Human Robot Interaction, ICRA, AAMAS, and IEEE RO-MAN, 2023 to 2026.
- **Teaching assistant.** Machine Learning and Reinforcement Learning, Cornell, 2023 to 2024. COS 126 Computer Science, Princeton, 2022 to 2023.